Euler equation for a perfect fluid

Consider a perfect fluid, flowing adiabatically through spacetime ($\zeta = \eta = \mathbf{B} = \mathbf{q} = 0$). In this case the Euler equations $\mathbf{h} \cdot (\nabla \cdot \mathbf{T}) = 0$ reduce to

$$(\rho + p)\,\mathbf{a} = -\mathbf{h} \cdot \nabla p\,. \tag{2.5.30}$$

In words:

$$\begin{pmatrix}\text{inertial mass per}\\ \text{unit volume}\end{pmatrix} \times (\text{4-acceleration}) = -\begin{pmatrix}\text{pressure gradient}\\ \text{projected orthogonal to } \mathbf{u}\end{pmatrix}.$$

Bernoulli equation

Consider a perfect fluid undergoing stationary, adiabatic flow in a stationary spacetime. More particularly, set $\zeta = \eta = \mathbf{B} = \mathbf{q} = 0$; assume that spacetime is endowed with a Killing vector field $\boldsymbol{\xi}$,

$$\xi_{\alpha;\beta} + \xi_{\beta;\alpha} = 0\,, \tag{2.5.31}$$

which need not be timelike; and assume that the flow is adiabatic, $ds/d\tau = 0$, and stationary in the sense that

$$\mathscr{L}_{\boldsymbol{\xi}}\, p = 0, \quad \nabla_{\boldsymbol{\xi}}\, \rho = \nabla_{\boldsymbol{\xi}}\, p = \cdots = 0\,. \tag{2.5.32}$$

Here $\mathscr{L}_{\boldsymbol{\xi}}$ is the Lie derivative along $\boldsymbol{\xi}$. In this case the Euler equations (2.5.30), together with the first law of thermodynamics (2.5.12), imply the relativistic Bernoulli equation

$$d(\mu\, \mathbf{u} \cdot \boldsymbol{\xi})/d\tau = 0\,; \tag{2.5.33}$$

i.e., $\mu\, \mathbf{u} \cdot \boldsymbol{\xi}$ is constant along flow lines.

Newtonian limit

The Newtonian limit of relativistic hydrodynamics is obtained when, in a nearly global Lorentz frame, the following approximations hold:

$$g_{00} = -(1 + 2\Phi), \quad |\Phi| \ll 1, \quad \mathbf{B}^2/\rho_0 \ll 1, \quad v^2 \ll 1\,. \tag{2.5.34}$$

$$p/\rho_0 \ll 1, \quad \Pi \ll 1, \quad v^2 \ll 1\,. \tag{2.5.35}$$

Here Φ is the Newtonian gravitational potential with sign $\Phi < 0$, and v is the ordinary velocity of the fluid

$$\mathbf{v} \equiv v_j = (u^j/u^0)\,\mathbf{e}_j\,. \tag{1}$$

In the Newtonian limit the chemical potential μ reduces to

$$\mu = m_B(1 + w)\,, \tag{2.5.36}$$

where w is the *enthalpy*

$$w = \Pi + p/\rho_0\,. \tag{2.5.37}$$

The Bernoulli equation (2.5.33) reduces to the familiar form

$$\Phi + \tfrac{1}{2}v^2 + w = \text{constant along flow lines}\,. \tag{2.5.38}$$

The Euler equation for a perfect fluid in adiabatic flow, (2.5.30), reduces to

$$\frac{d\mathbf{v}}{dt} = -\nabla\Phi - \frac{1}{\rho_0}\nabla p\,, \tag{2.5.39}$$

where d/dt, the derivative with respect to proper time along the flow lines, has the Newtonian form

$$d/dt = \partial/\partial t + \mathbf{v} \cdot \boldsymbol{\nabla}\,. \tag{2.5.40}$$

The first law of thermodynamics, (2.5.12) and (2.5.12′), reduces to

$$d\Pi = -p\,dV_0 + T\,ds_0\,, \tag{2.5.41}$$

$$dw = V_0\,dp + T\,ds_0\,. \tag{2.5.42}$$

(In Newtonian theory one usually adopts a per-unit-mass viewpoint rather than a per-baryon viewpoint; and thus one uses ρ_0, V_0 rather than n, $\mathcal{V}$, s_0.)

A. Radiative Transfer

Basic references Appendix 1 of Pacholczyk (1970); Mihalas (1970); Chandrasekhar (1960); Appendix 1 of Pacholczyk (1970); Mihalas (1970).

Notation and terminology

At an arbitrary event in spacetime pick an arbitrary local Lorentz frame. In that frame pick an arbitrary spatial direction $\mathbf{n}$ (a unit vector; see Figure 2.6.1). Examine the amount of energy dE that is (i) carried by photons across a unit surface area dA orthogonal to $\mathbf{n}$ (the surface $\mathscr{S}_m$ of Figure 2.6.1) during unit time dt, with (ii) the photons having frequencies ν in the range $d\nu$, and (iii) the photons being directed into a solid angle $d\Omega$ about $\mathbf{n}$. The ratio

$$I_\nu = \frac{dE}{dt\; dA\; d\nu\; d\Omega} \tag{2.6.1}$$

is called the *specific intensity* or simply the *intensity*. It depends on (a) the event in spacetime; (b) the choice of Lorentz frame; (c) the direction $\mathbf{n}$; (d) the frequency ν. One also defines the *total intensity* I by

$$I = \int I_\nu \, d\nu = \frac{dE}{dt\, dA\, d\Omega}\,; \tag{2.6.2}$$

the *average specific intensity* or *average intensity* J_ν (averaged over all directions) by

$$J_\nu = \frac{1}{4\pi} \int I_\nu \, d\Omega\,; \tag{2.6.3}$$

and the *average total intensity* J by

$$J = \frac{1}{4\pi} \int I \, d\Omega = \frac{1}{4\pi} \int I_\nu \, d\nu \, d\Omega\,. \tag{2.6.4}$$

It is easy to see that the *energy density per unit frequency* in the radiation is

$$\rho^{(\mathrm{rad})} = \frac{dE}{d^3x\, d\nu} = \frac{4\pi J_\nu}{c}\,, \tag{2.6.5}$$

and that the *total energy density* in the radiation is

$$\rho^{(\mathrm{rad})} = \frac{dE}{d^3x} = \frac{4\pi J}{c}\,. \tag{2.6.6}$$

Pick a 2-surface $\mathscr{S}_m$ in the chosen Lorentz frame, with unit normal $\mathbf{m}$ (Fig. 2.5.1). Examine the total energy dE per unit area dA that is carried across the chosen surface during unit time dt, by photons having frequency ν in the range $d\nu$. Make no restrictions on the photon direction or solid angle; but count negatively those photons that cross dA from the front side toward the back. The ratio

$$F_\nu = \frac{dE}{dt\, dA\, d\nu} \tag{2.6.7}$$

is called the *specific flux*, or simply the *flux*. It is easy to see from Figure 2.5.1 that

$$F_\nu = \int I_\nu \cos\theta \, d\Omega\,. \tag{2.6.8}$$

[Figure 2.6.1 placeholder]
The surfaces, directions, and solid angle used in the definition of intensity and of flux. $\mathscr{S}_m$ is a surface with unit normal $\mathbf{m}$; $\mathbf{n}$ is a unit vector in the photon propagation direction; θ is the angle between $\mathbf{m}$ and $\mathbf{n}$; $d\Omega$ is the solid angle element.

FIG. 1. The surfaces, directions, and solid angle used in the definition of intensity and of flux.

The specific flux depends on (a) the chosen event in spacetime; (b) the chosen Lorentz frame at that event; (c) the chosen 2-surface $\mathscr{S}_m$ in that Lorentz frame.

The integral of the flux over all frequencies is called the *total flux*

$$F = \int F_\nu \, d\nu = \int I \cos\theta \, d\Omega\,. \tag{2.6.9}$$

Notice that the total flux is the “first moment” of the total intensity (the “moment” being taken with respect to the unit normal $\mathbf{m}$). One can readily verify that the “second moment” is equal to the radiation pressure that acts across the surface $\mathscr{S}_m$:

$$T^{(\mathrm{rad})} \equiv \mathbf{m} \cdot \mathbf{T}^{(\mathrm{rad})} \cdot \mathbf{m} = \int I \cos^2\theta \, d\Omega\,. \tag{2.6.10}$$

Invariance of I_ν/ν^3

Choose a particular photon that passes through a given event in spacetime. Observe that particular photon, and all others in the vicinity, from several different Lorentz frames at the given event. The frequency ν of the photon will depend on the choice of Lorentz frame (Doppler shift from one frame to another). The specific intensity I_ν in the frame neighborhood of the photon will also depend on the choice of Lorentz frame. However, the ratio I_ν/ν^3 will be Lorentz- invariant. (Aside from a factor h^{-4}, I_ν/ν^3 is the invariant number density in phase space

$$\mathscr{N} = \frac{dN}{d^3x\, d^3p} = \frac{1}{h^3}\,\frac{I_\nu}{\nu^3}\,; \tag{2.6.11}$$

see, e.g., §22.6 of MTW.)

When photons propagate freely through curved spacetime (no interaction with matter), the ratio I_ν/ν^3 is conserved along the world line of each photon (Liouville's theorem).